

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1
PAPER 1 June 2023
2 hours 30 minutes

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Evaluate $8 + 6 \times 3 - 4$. [2]
2. Evaluate $-12 - (-6)$. [3]
3. Work out $\frac{1}{3} + 1$. Give your answer as a mixed number if possible. [3]
4. Find 40% of 40. [3]
5. Solve $6x + 3 = 51$. [3]
6. Share \$21 in the ratio 2 : 5. [4]
7. Find (a) the HCF and (b) the LCM of 12 and 42. [4]
8. Write 2000 in standard form. [3]
9. Simplify $3^3 \times 3^4$. Leave your answer as a power of 3. [3]
10. $n(A) = 12$, $n(B) = 10$ and $n(A \cap B) = 5$. Find $n(A \cup B)$. [3]
11. Expand and simplify $2(x + 4) - 1x$. [3]
12. Factorise completely $30x + 10$. [3]
13. Factorise $x^2 + 11x + 30$. [4]
14. Make h the subject of the formula $A = 2\pi r(r + h)$. Take π as a symbol. [4]
15. Solve $3x + 4 < 10$. [3]
16. Solve the simultaneous equations
$$\begin{aligned}x + y &= 8 \\x - y &= 4.\end{aligned}$$
[4]
17. The n th term of a sequence is $4 + 4(n - 1)$. Find the 8th term. [3]
18. In triangle ABC, angle $A = 48^\circ$ and angle $B = 64^\circ$. Find angle C. [3]
19. A regular polygon has 8 sides. Find the size of each exterior angle. [3]
20. A right-angled triangle has shorter sides 9 cm and 12 cm. Calculate the hypotenuse. [4]
21. In right-angled triangle XYZ, angle $X = 60^\circ$ and hypotenuse $XZ = 8$ cm. Calculate XY, the side adjacent to 60° . [4]

- 22.** A triangle has base 6 cm and perpendicular height 7 cm. Find its area. **[3]**
- 23.** Calculate the area of a circle of radius 21 cm. Take $\pi = 22/7$. **[4]**
- 24.** A cylinder has radius 7 cm and height 3 cm. Find its volume. Take $\pi = 22/7$. **[4]**
- 25.** Find the gradient of the line joining (2, 6) and (4, 8). **[3]**
- 26.** Find the midpoint of the line joining (2, 6) and (4, 8). **[3]**
- 27.** A bag contains 4 red, 2 blue and 2 green beads. A bead is drawn at random. Find the probability that it is red. **[3]**
- 28.** Find the mean of 10, 11, 12, 9, 12. **[3]**
- 29.** Work out $\begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix}$ where $\begin{pmatrix} & \\ & \end{pmatrix}$ denotes a 2×2 matrix times a column vector. **[4]**
- 30.** A trader in Mbare buys a crate of tomatoes for \$20 and sells it for \$32. Calculate the percentage profit. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).